

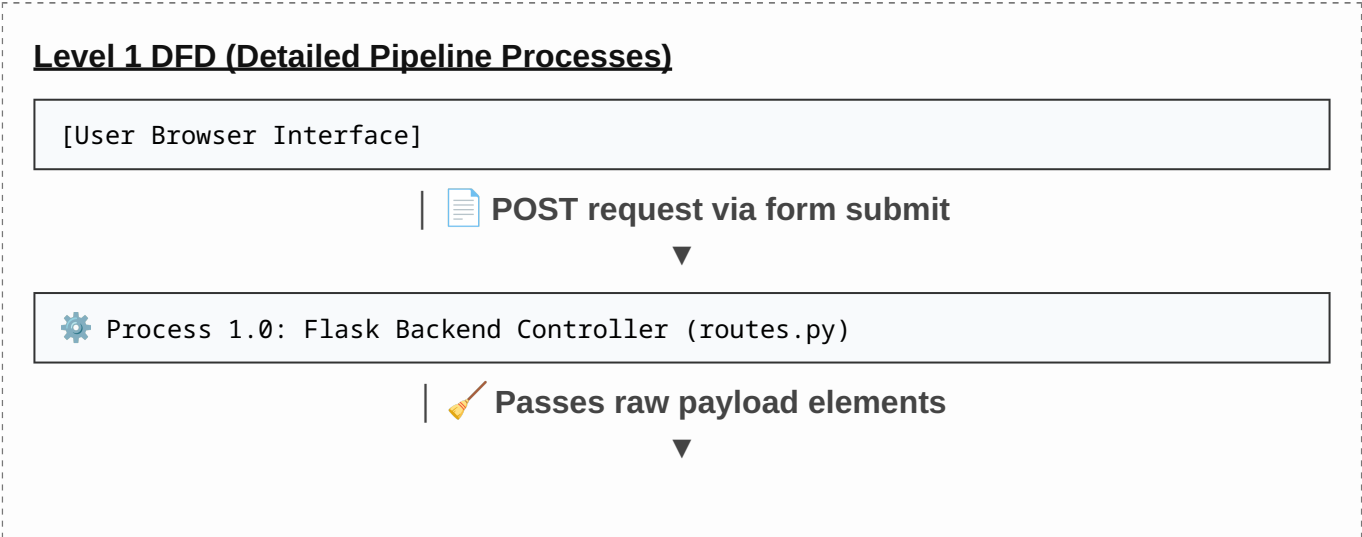
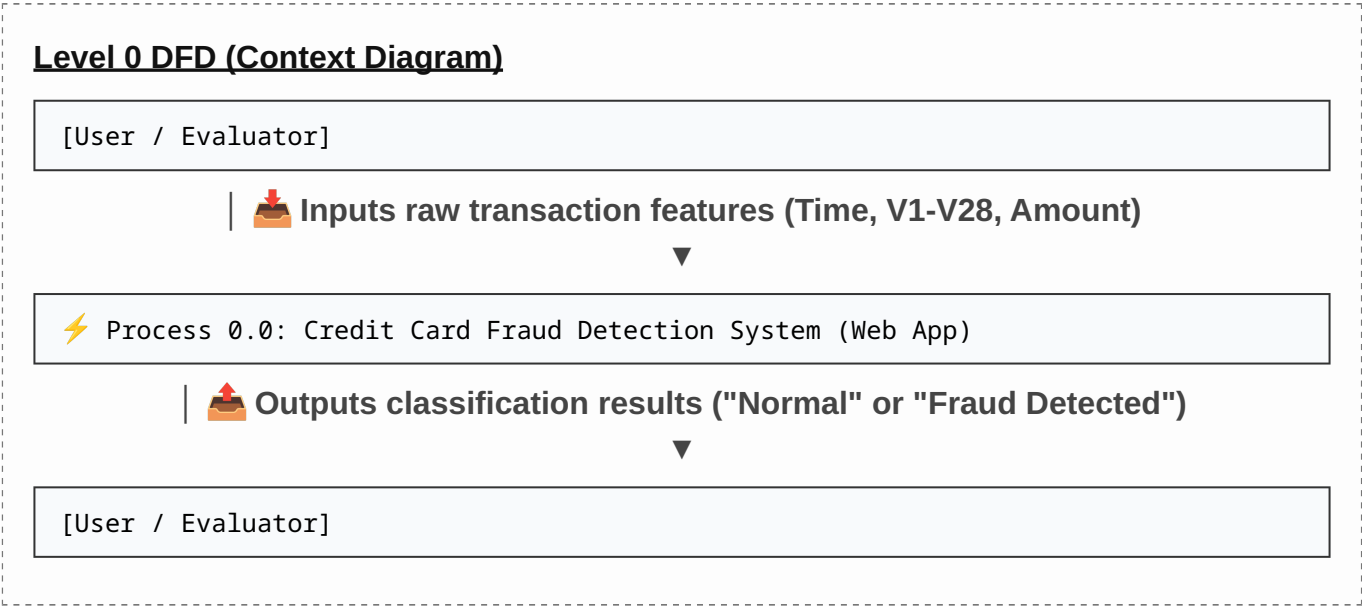
Project Design Phase-II

Data Flow Diagram & User Stories

Date	6 july 2026
Team ID	XXXXXX
Project Name	Credit Card Fraud Detection System using Machine Learning
Maximum Marks	4 Marks

1. DATA FLOW DIAGRAM (DFD)

Below is the structured functional flow of data through the machine learning pipeline, from the frontend interface to model inference and backend evaluation.



⚙️ Process 2.0: Data Preprocessing Engine (StandardScaler Transformation)

|  Transformed Feature Array (Shape: 1x30)



⚙️ Process 3.0: Model Inference Engine (Loads.pkl via Joblib)

|  Requests trained parameters



 Data Store D1: Saved Machine Learning Model (XGBoost / Random Forest)

|  Binary Prediction Output (0 or 1)



⚙️ Process 4.0: Response Renderer Engine

|  UI payload rendering ("Normal Transaction" / "Fraud Transaction Detected")



[User Browser Interface]

2. CORE AGILE USER STORIES

- **User Story 1:** As a bank compliance evaluator, I want to submit raw numeric PCA features (\$V1\$ to \$V28\$) alongside transaction amount and time so that I can immediately verify if a transaction shows high-risk fraudulent anomalies without human manual review.
- **User Story 2:** As a project reviewer, I want to access the application via a public Render URL without local environment setups so that I can rapidly audit the runtime behavior and verify live prediction speeds.
- **User Story 3:** As a system developer, I want to inspect the complete version-controlled code, EDA visuals, and evaluation metrics on GitHub so that I can reproduce, scale, or integrate the pipeline into core banking microservices.